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METHOD TO PROMOTE AIRCRAFT ELECTRIC POWER GENERATING SYSTEM FAULTS FROM LOWER SEVERITY TO HIGHER SEVERITY BASED ON FREQUENCY OF OCCURRENCE

Abstract

A fault analysis system is disclosed. The fault analysis system includes one or more processors configured to: continuously capture status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; count a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; generate a second fault event associated with the fault event based on comparing the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and report the second fault event according to the second severity level.

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Background/Summary

BACKGROUND

[0001] The subject matter disclosed herein generally relates to fault events in an aircraft electric power generating system (EPGS), and more particularly to promotion of aircraft EPGS fault events from lower severity to higher severity based on frequency of occurrence.

[0002] In an aircraft EPGS, system fault events can be reported to the aircraft crew based on detected events. Techniques supportive of effective reporting are desired.

BRIEF DESCRIPTION

[0003] A fault analysis system is disclosed, including: one or more processors configured to: continuously capture status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; count a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; generate a second fault event associated with the fault event based on comparing the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and report the second fault event according to the second severity level.

[0004] Any one or combination of the foregoing embodiments, further including: a memory device including a circular buffer, wherein the one or more processors are configured to continuously store the status signals to the circular buffer.

[0005] Any one or combination of the foregoing embodiments, wherein the one or more processors are configured to classify the plurality of fault events according to respective severity levels of the plurality of fault events.

[0006] Any one or combination of the foregoing embodiments, wherein the one or more processors are further configured to: add the second fault event to the plurality of fault events; order the plurality of fault events based on respective severity levels of the plurality of fault events; and display the plurality of fault events based on the order.

[0007] Any one or combination of the foregoing embodiments, wherein: the target severity level includes a lowest severity level; and the second severity level is higher than the target severity level.

[0008] Any one or combination of the foregoing embodiments, further including: one or more rule based tables, wherein the one or more processors are configured to generate the second fault event based on the fault event and the one or more rule based tables.

[0009] Any one or combination of the foregoing embodiments, wherein: the one or more processors are configured to assign the second severity level to the second fault event.

[0010] Any one or combination of the foregoing embodiments, wherein: the one or more processors are configured to generate a database entry corresponding to the fault event and store the database entry to a database; and the one or more processors are configured to generate a second database entry corresponding to the second fault event and store the second database entry to the database.

[0011] Any one or combination of the foregoing embodiments, wherein: the one or more processors are configured to generate first timestamp data associated with the fault event and second timestamp data associated with generation of the second fault event; and the one or more processors are configured to display the fault event and the second fault event based on the first timestamp data and the second timestamp data.

[0012] Any one or combination of the foregoing embodiments, wherein the one or more processors are further configured to: generate one or more second corrective actions associated with the second fault event, wherein at least a portion of the one or more second corrective actions associated with the second fault event is different from one or more correction actions associated with the fault event.

[0013] Any one or combination of the foregoing embodiments, wherein the one or more processors are further configured to: determine a pattern associated with the instances of the fault event; and generate the second fault event based on comparing the pattern to a target pattern.

[0014] An apparatus is disclosed, including: a controller including one or more processors and logic circuitry, wherein: processing circuitry of the one or more processors is configured to continuously capture status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; the logic circuitry is configured to count a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; the processing circuitry is configured to generate a second fault event associated with the fault event based on comparing, by the logic circuitry, the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and the processing circuitry is configured to report the second fault event according to the second severity level.

[0015] Any one or combination of the foregoing embodiments, further including: a memory device including a circular buffer, wherein the processing circuitry is configured to continuously store the status signals to the circular buffer.

[0016] Any one or combination of the foregoing embodiments, wherein the logic circuitry is configured to classify the plurality of fault events according to respective severity levels of the plurality of fault events.

[0017] Any one or combination of the foregoing embodiments, wherein the processing circuitry is configured to: add the second fault event to the plurality of fault events; order the plurality of fault events based on respective severity levels of the plurality of fault events; and display the plurality of fault events based on the order.

[0018] A method is disclosed, including: continuously capturing, by one or more circuitry, status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; counting, by the one or more circuitry, a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; generating, by the one or more circuitry, a second fault event associated with the fault event based on comparing the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and reporting, by the one or more circuitry, the second fault event according to the second severity level.

[0019] Any one or combination of the foregoing embodiments, further including: continuously storing, by the one or more circuitry, the status signals to a circular buffer.

[0020] Any one or combination of the foregoing embodiments, further including: classifying, by the one or more circuitry, the plurality of fault events according to respective severity levels of the plurality of fault events.

[0021] Any one or combination of the foregoing embodiments, further including: adding, by the one or more circuitry, the second fault event to the plurality of fault events; ordering, by the one or more circuitry, the plurality of fault events based on respective severity levels of the plurality of fault events; and displaying, by the one or more circuitry, the plurality of fault events based on the ordering.

[0022] Any one or combination of the foregoing embodiments, wherein: the target severity level includes a lowest severity level; and the second severity level is higher than the target severity level.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike.

[0024] FIG. **1** is a schematic illustration of a system that can incorporate various embodiments of the present disclosure.

[0025] FIG. **2** is an example flowchart supportive of promotion of aircraft EPGS fault events from lower severity to higher severity in accordance with example aspects of the present disclosure.

[0026] FIG. **3** illustrates an example flowchart of a method that supports promotion of aircraft EPGS fault events from lower severity to higher severity in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0027] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

[0028] FIG. **1** is a schematic illustration of a system **100** that can incorporate various embodiments of the present disclosure. As shown in FIG. **1**, the system **100** may include an aircraft **101** and one or more computing devices **103**.

[0029] According to one or more embodiments of the present disclosure, the system **100** may support fault reporting and fault analysis by the aircraft system **102**. The system **100** may support fault reporting and fault analysis by a computing device **103-***a* or computing device **103-***b* (e.g., based on data provided by the aircraft system **102**).

[0030] The computing devices **103** may be capable of performing any combination of operations described herein. In some cases, the system **100** may include computing devices **103** internal to and integrated with the aircraft system **102** and computing devices **103** separate from the aircraft system **102**. Components of the aircraft system **102** described herein may be electrically coupled via any combination of buses (not illustrated) (e.g., power buses, data buses, avionics buses, or the like).

[0031] A computing device **103** (e.g., computing device **103-***a*, computing device **103-***b*) may be disposed in operable communication with components of the aircraft **101**. The system **100** and aircraft system **102** supports communication between the computing devices **103** and other devices of the system **100** via wired communication protocols, wireless communication protocols (e.g., electromagnetic (EM) signals, WiFi, Bluetooth™, ZigBee™, Ubiquiti™, 3G, 4G, LTE, and the like), and/or combinations including one or more of the foregoing.

[0032] The computing device **103** is configured to receive, store and/or transmit data. The computing device **103** includes processing components configured to analyze received signals and data described herein. The computing device **103** includes processing components configured to provide data (and/or control signals) to other components of the system **100** or aircraft system **102**. The computing device **103** includes any number of suitable components, such as processors, memory, communication devices and power sources.

[0033] The computing device **103** may include processing circuitry capable of executing instructions stored on a memory of the computing device **103** in association with performing one or more functions described herein. Some elements stored in the memory may be described as or referred to as instructions or instruction sets, and some functions of the computing device **103** may be implemented using machine learning techniques.

[0034] The aircraft **101** can include an EPGS **105**. The EPGS **105** may include an EPGS controller

110. In some examples, logic circuitry **115-*a*** may be implemented in the EPGS controller **110**. The logic circuitry **115-*a*** may be, for example, built-in test equipment (BITE) logic. In some embodiments, the EPGS **105** may include multiple EPGS controllers **110**. It is to be understood that aspects described herein with reference to a EPGS controller **110** may be applied to other EPGS controllers **110** included in the EPGS **105**.

[0035] The EPGS controller **110** may include microcontroller **116** (or multiple microcontrollers **116**). Each of the EPGS controller **110** and the microcontroller **116** may correspond to one or many computer processing devices. For example, may include a silicon chip, such as a FPGA, an ASIC, any other type of IC chip, a collection of IC chips, or the like. In some aspects, the processors may include a microprocessor, CPU, a GPU, or plurality of microprocessors configured to execute the instructions sets stored in a corresponding memory (e.g., memory **117** of the microcontroller **116**). For example, upon executing the instruction sets stored in memory **117**, the microcontroller **116** may enable or perform one or more functions described herein. Some elements stored in memory **117** may be described as or referred to as instructions or instruction sets.

[0036] The memory **117** may include one or multiple computer memory devices. The memory **117** may include, for example, Random Access Memory (RAM) devices, Read Only Memory (ROM) devices, flash memory devices, magnetic disk storage media, optical storage media, solid state storage devices, core memory, buffer memory devices, combinations thereof, and the like. The memory **117**, in some examples, may correspond to a computer readable storage media. In some aspects, the memory **117** may be internal or external to the EPGS controller **110** and/or the microcontroller **116**.

[0037] The memory **117** may be configured to store instruction sets and other data structures (e.g., depicted herein) in addition to temporarily storing data for the microcontroller **116** to execute various types of routines or functions. For example, the memory **117** may be configured to store program instructions (instruction sets) that are executable by the microcontroller **116** and provide functionality of the EPGS controller **110** described herein. The memory **117** may also be configured to store data or information that is useable or capable of being called by the instructions stored in memory **117**.

[0038] The aircraft system **102** may include a database **104**. The aircraft system **102** supports storing and/or accessing data described herein (e.g., status signals **112**, fault-confirmation signals **114**, fault events **121**, status information **119**, fault records **137**, fault analysis data **145**, and the like, example aspects of which will be described herein) via the database **104**.

[0039] The EPGS controller **110** may include logic circuitry **115-*b*** configured to promote respective severity levels of one or more fault events (also referred to herein as faults), example aspects of which as described herein. The logic circuitry **115-*b*** may be referred to herein as promotion logic.

[0040] In some embodiments, the logic circuitry **115-*b*** logic may be provided by the EPGS controller **110** through a data-only configuration file (e.g., a parameter data item). The data-only configuration file may support flexibility in the tuning of the logic circuitry **115-*b*** without impacting the certification baseline of the EPGS controls. The aircraft system **102** supports updating the logic circuitry **115-*b*** via one or more data files (not illustrated). For example, the aircraft system **102** supports providing a data file update to the EPGS controls based on an update schedule, for example, in a manner similar to avionics navigation database monthly updates.

[0041] In some examples, the data file may include one or more tables of configuration information for the logic circuitry **115-*b***. In some other examples, the data file may include a fault catalog including correlations between faults and associated causes (e.g., performance parameters of an LRU **120**, status signals **112** associated with an LRU **120**, or the like). In some examples, the fault catalog may be implemented in one or more rule based tables **113**.

[0042] The aircraft **101** may include multiple line replaceable units (LRUs) **120** electrically coupled to the EPGS **105**. The LRUs **120** may be coupled to the EPGS **105** via one or more data buses. Non-limiting examples of the LRUs **120** include generators, contactors, controllers, switches, wiring, sensors, and the like.

[0043] The EPGS **105** may include a non-volatile memory (NVM) device **125** capable of storing data. Aspects of the present disclosure include implementing a circular buffer **130** in the NVM device **125**. The circular buffer **130** (also referred to herein as a circular queue, a cyclic buffer, or a ring buffer) is a data structure that uses a single buffer as if the buffer were connected end-to-end. The size of the circular buffer **130** may be a fixed size or may be tunable by the system **100**.

[0044] The circular buffer **130** may have a predetermined storage space. In some cases, the circular buffer **130** may be capable of continuously storing data. For example, a property of the circular buffer **130** is that when the circular buffer **130** is full and a subsequent writing of data to the circular buffer **130** is performed, the data write may overwrite the oldest data stored on the circular buffer **130**. In some aspects, the techniques described herein include setting the storage space of the circular buffer **130** based on one or more criteria.

[0045] According to one or more embodiments of the present disclosure, the system **100** supports promoting the severity of fault events associated with the aircraft **101**. The techniques described support reclassifying fault events (e.g., as being relatively less severe to relatively more severe) based on one or more criteria. The reclassifying of fault events may support improving accuracy, efficiency, and predictive analysis associated with system troubleshooting of the aircraft **101**. References herein to reclassifying fault events may include the promotion of fault events and the generation of additional fault events (e.g., of higher severity) from existing fault events (e.g., of relatively less severity), example aspects of which are described herein.

[0046] The terms ‘faults’, ‘fault events’, ‘system faults,’ and ‘system fault events’ may be used interchangeably herein. In some examples, the terms may refer to faults associated with the aircraft system **102**. For example, in some cases, the terms may refer to faults associated with the EPGS **105** and LRUs **120**. Descriptions with reference to the ‘faults’, ‘fault events’, and ‘system faults’ may include occurrences of a fault (e.g., generator overvoltage), reporting of the fault (e.g., by the EPGS **105**, the aircraft system **102**, or the like), and/or generation of an additional fault (e.g., based on promotion or reclassification of an existing fault as described herein).

[0047] In some aspects, the aircraft system **102** supports reporting of fault events at different levels to the aircraft crew (e.g., pilots, maintenance crew, and the like) based on severity of the event being detected. For example, in response to a fault event of generator overvoltage, the aircraft system **102** (e.g., at EPGS **105**) may remove the generating channel from the aircraft network in order to protect the aircraft **101** from the fault event. In an example, the aircraft system **102** (e.g., at computing device **103-*a***) may generate and output a notification **118** (e.g., an audible, visible, and/or haptic notification via one or more displays or computing devices **103**) associated with the fault event. The notification **118** may be text-based, graphics based, or the like. For example, the aircraft system **102** may generate and output a notification **118** to the cockpit as a pilot crew warning action (e.g., ‘Electric Generator Offline’).

[0048] The aircraft system **102** supports monitoring and reporting of fault events of different levels of severity. For example, the aircraft system **102** supports reporting fault events with a notification **118** for additional pilot crew workload to monitor and/or address the fault event. For example, for a fault event of relatively less severity (e.g., an elevated generator oil temperature), the aircraft system **102** may generate and output a corresponding notification **118** (e.g., ‘Oil Temperature Advisory’).

[0049] In some aspects, the aircraft system **102** supports providing status information **119** (e.g., ‘Generator Low Oil Level’), which in some cases, may be more relevant to the maintenance crew than to the pilot crew. The EPGS **105** is capable of communicating data (e.g., status signals **112** (later described herein), fault-confirmation signals **114** (later described herein), notifications **118**, status information **119**, and the like) indicative of warning/advisory/status events within the aircraft system **102**. In some aspects, the EPGS **105** is capable of storing data associated with the

warning/advisory/status events in NVM device **125** as fault records **137** usable by the system **100** (or an operator of the system **100**) for troubleshooting of the aircraft system **102**.
[0050] According to one or more embodiments of the present disclosure, in response to a fault event at the aircraft **101** (e.g., at a LRU **120**), the EPGS controller **110** may isolate the reason for the fault event to the most relevant LRU **120** in the aircraft system **102**. In some examples, the fault event may be due to one or more fault conditions, example aspects of which are later described herein.
[0051] In an example, circuitry in the EPGS controller **110** may provide control signals **111** to microcontroller **116** to consume. For example, the EPGS controller **110** provides, in digital form (e.g., control signals **111**), conditions of switches and measurement data (e.g., temperatures, pressures, and the like) to microcontroller **116** to consume. Based on the control signals **111**, the microcontroller **116** may produce status signals **112**.
[0052] The EPGS controller **110** (processor) may act on the status signals **112** to determine whether a fault condition is present. For example, the EPGS controller **110** (e.g., using logic circuitry **115**-*a*) may generate and output a fault-confirmation signal **114** indicating the presence of the fault condition. That is, for example, the EPGS controller **110** may generate a fault code (e.g., a fault-confirmation signal **114**) for confirming the fault event. In an example, a fault event may include a generator failure associated with the aircraft system **102**, but is not limited thereto.
[0053] Non-limiting examples of the status signals **112** include voltages, currents, frequencies, temperatures, internal controller statuses, switches, contactors, and the like.
[0054] The aircraft system **102** provides increased effectiveness compared to other techniques for reporting of fault events by an EPGS. For example, some other techniques for EPGS reporting may collect and store events/faults by flight legs, with little to no correlation of faults across flight legs.
[0055] According to one or more embodiments of the present disclosure, the EPGS **105** is configured to apply additional logic equations (e.g., implemented by logic circuitry **115**-*b*) by searching fault records **137** and promoting faults (e.g., lower-level faults, mid-level faults, or the like) to a greater severity level based on a frequency of occurrence of the faults with respect to a target quantity of flight legs (and/or with respect to a time period). In some embodiments, the EPGS **105** may search the fault records **137** in response to one or more trigger criteria.
[0056] Non-limiting examples of the trigger criteria include any one or a combination of a threshold temporal condition (e.g., every 12 hours, every day, every week, or the like), a threshold quantity of flights (e.g., every 2 flights, every 4 flights, or the like), and a user input. Accordingly, for example, the EPGS **105** may autonomously and/or semi-autonomously perform the techniques described herein.
[0057] In some aspects, the logic circuitry **115**-*b* is configured to count the quantity of instances of faults over a given number of flight legs. For example, the EPGS **105** (e.g., using the logic circuitry **115**-*b*) may determine, from any quantity of flight leg data **140**, the quantity of instances of faults over a given number of flight legs. In an example, each instance of flight leg data **140** may be associated with a flight leg (e.g., a flight originating at a first location and ending at a second location).
[0058] Among the faults, the EPGS **105** (e.g., using the logic circuitry **115**-*b*) may analyze any faults of a severity level less than or equal to a target severity level. Non-limiting examples of the target severity level may include low severity (e.g., a low-level fault), medium severity (e.g., a mid-level fault), high severity (e.g., a high-level fault), and severity levels between any of the low, medium, or high severity levels. In some aspects, the system **100** supports using a scoring metric (e.g., a numerical score) for indicating severity level.
[0059] In response to identifying that a target fault of a given severity level (e.g., low-level fault) exceeds a frequency of occurrence with respect to a target quantity of flight legs (and/or with respect to a time period), the EPGS **105** may promote the target fault event to a relatively higher severity level. For example, the EPGS **105** may assign a higher severity level to the fault event.

[0060] In some aspects, the EPGS **105** may report the fault as a new fault and include the new fault in the fault records **137**. That is, for example, based on the fault, the EPGS **105** (e.g., by controller **110**) may generate the new fault. Accordingly, for example, the aircraft system **102** (e.g., via computing device **103**-*a*) may list or display the new fault among faults to be resolved, at a higher priority position compared to the original fault, example aspects of which will be described herein. In some examples, the EPGS **105** may apply the logic equations specific to consider target requirements of the EPGS **105** for a given aircraft type, and may include specific logic considerations from aircraft operator input. It is to be understood that descriptions of operations performed by the aircraft system **102** may include performance of the operations by one or more components (e.g., computing device **103**-*a*, EPGS **105**, controller **110**, and the like) included in the aircraft system **102**.
[0061] In an example implementation, the EPGS **105** may have identified and reported a relatively low severity fault (e.g., generator oil level status: low oil) every 3 or 4 flights. For example, among flight leg data **140**-*a* through flight leg data **140**-*n* (where each instance of flight leg data **140** corresponds to a completed flight), every 3rd or 4th instance of flight leg data **140** may include the same relatively low severity fault.
[0062] The EPGS **105** (e.g., using logic circuitry **115**-*b*) may compare the quantity of the relatively low severity fault to a threshold quantity of instances. In response to a comparison result in which the quantity is greater than or equal to the threshold quantity of instances, the EPGS **105** (e.g., using the logic circuitry **115**-*b*) may promote the relatively low severity fault (e.g., assign a higher severity level to the fault). For example, the EPGS **105** may generate and report a fault of the higher severity level (also referred to herein as a higher severity fault).
[0063] In some examples, reporting the higher severity fault may include displaying the fault according to a higher priority compared to faults of a relatively lower severity level. In some examples, reporting the higher severity fault may include displaying (e.g., on a display screen) the higher severity fault in combination with one or more visual indicators (e.g., color highlighting, bold text) representative of the higher severity level. In some examples, the one or more visual indicators may be representative of the change in severity level (e.g., the promotion to the higher severity level) for the fault. For example, the one or more visual indicators may indicate that the higher severity fault has been newly generated based on a different fault (e.g., the lower severity fault).
[0064] In the example implementation, for the higher severity fault, the aircraft system **102** may generate or display a notification **118** including an advisory or warning level message ‘Persistent Generator Cooling System Oil Leak’. In some cases, though addressing the persistent generator cooling system oil leak through repairs and/or equipment replacement may involve a relatively larger amount of maintenance action compared to other actions (e.g., adding oil to the generator cooling system), the overall cost associated with the repairs and/or equipment replacement may be less than the overall cost associated with repeatedly performing actions for addressing the related low severity fault. For example, repair and/or replacement of the generator cooling system may be cumulatively less time-consuming and less expensive (e.g., due to a resulting reduction in maintenance hours and/or a reduction in consumables (oil)) compared to having a maintenance crew service the generator cooling system and replace the oil every 3 to 4 flights.
[0065] As an example operational flow, the aircraft system **102** generates a notification **118** instructing the crew to troubleshoot the generator cooling system. The aircraft system **102** may present the notification **118** via a computing device **103**-*a* onboard the aircraft system **102** and/or a computing device **103**-*b* (e.g., a portable computing device, a workstation, or the like) separate from the aircraft system **102**. In some examples, the aircraft system **102** may transmit data corresponding to the higher severity fault and the notification **118** to the computing device **103**-*b*, and the computing device **103**-*b* may display the notification **118**.
[0066] Based on the troubleshooting, the crew finds a leak in a cooling system fitting. The crew

performs the associated actions to resolve the leak. In response to the resolution of the leak, the aircraft system **102** may reset the notification **118** in the EPGS controls to complete the action. In some examples, the aircraft system **102** may reset the notification **118** in response to a user input confirming that the actions have been completed. Additionally, or alternatively, the aircraft system **102** may reset the notification **118** in response to detecting through sensor data or the status signals **112** that the leak has been resolved.

[0067] The systems and techniques described herein provide improvements compared to other systems and approaches for attending to faults in an aircraft **101**. For example, for the example case of the relatively low severity fault, some other systems and approaches merely include servicing the low severity fault by a maintenance crew (e.g., the maintenance crew services the generator oil level by adding oil) each time the fault is reported to address the issue. Accordingly, for example, by failing to analyze the occurrence frequency of the relatively low severity fault and failing to promote the severity, such other systems and approaches may accordingly fail to recognize or suggest alternative and/or additional actions (e.g., repair and/or replacement of equipment) which may otherwise result in a reduced overall cost and increased safety.

[0068] Further, for example, by failing to analyze the occurrence frequency of the relatively low severity fault and failing to promote the severity, such other systems and approaches may accordingly fail to effectively troubleshoot and identify a larger or hidden fault (e.g., leak in a cooling system fitting) that is causing or is associated with the relatively low severity fault. For example, the systems and techniques described herein may support identifying, from relatively low severity faults (e.g., bolts which are reported as loose every 3 to 4 flights and are then tightened by maintenance crew), a related larger issue or defect to be addressed.

[0069] The systems and techniques described herein further support increased efficiency through enhanced troubleshooting of faults and identifying potential larger or hidden faults. For example, some other approaches may fail to automatically recognize that a relatively low severity fault is being caused by a larger or hidden fault. For example, some other approaches may rely on manual review of a maintenance logbook or records by maintenance staff, which is more time intensive, less efficient, and less accurate compared to the techniques described herein.

[0070] In some embodiments, the aircraft system **102** may provide the fault records **137** to a computing device **103** (e.g., computing device **103**-*a*, computing device **103**-*b*, or the like). Based on processing the fault records **137**, the computing device **103** may generate fault analysis data **145** including the notification **118** described herein. In some embodiments, the computing device **103** may promote the relatively low severity fault and/or identify the larger or hidden fault based on processing the fault records **137**.

[0071] In some embodiments, the fault analysis data **145** may include a correlation between the relatively low severity fault and the identified larger or hidden fault. In some embodiments, using the fault analysis data **145**, the system **100** may support determining a correlation between relatively high severity faults and candidate faults of lower severity. Accordingly, for example, the system **100** may support effective determination and identification of potentially costly high severity faults from repeated instances of relatively lower severity faults. Accordingly, for example, the techniques described herein provide more effective and more accurate insight of fault events compared to other techniques which fail to incorporate promotion and prioritization of lower severity faults as described herein.

[0072] FIG. **2** illustrates an example flowchart of a method **200** in accordance with one or more embodiments of the present disclosure. The method **200** may be implemented by aircraft system **102** (e.g., at EPGS **105**) and/or a computing device **103** described with reference to FIG. **1**.

[0073] At **205**, the method **200** may include capturing status signals **112** corresponding to fault events **121**. In some examples, the method **200** may include continuously capturing status signals **112** corresponding to fault events **121**. In an example, the fault events **121** are associated with the EPGS **105** and/or LRUs **120** as described herein. In some aspects, the EPGS **105** (e.g., controller

110) is configured to continuously store the status signals **112** to the circular buffer **130**.

[0074] In an example, the circular buffer **130** may be capable of storing status signals **112** for a duration equal to a temporal period (e.g., based on available memory of the circular buffer **130**). When the circular buffer **130** is full and a subsequent writing of data to the circular buffer **130** is performed, the data write may overwrite the oldest data stored on the circular buffer **130**.

[0075] At **207**, the method **200** may include classifying the fault events **121** according to respective severity levels of the fault events **121**. For example, in some embodiments, the EPGS **105** (e.g., using logic circuitry **115**-*a* and/or the logic circuitry **115**-*b*) is configured to classify the fault events **121** according to respective severity levels (e.g., low severity, medium severity, high severity, or the like) of the fault events **121**. In some aspects, the EPGS **105** (e.g., using logic circuitry **115**-*a* and/or the logic circuitry **115**-*b*) is configured to classify the fault events **121** using a scoring metric for indicating the severity levels.

[0076] At **210**, the method **200** may include searching the circular buffer **130** for status signals **112**. In some embodiments, the method **200** may include searching the circular buffer **130** based on a target temporal period (e.g., the past X hours, the past X days, the past X weeks, or the like). In some embodiments, the system **100** is configured to set or tune the target temporal period. In some embodiments, the method **200** may include searching the circular buffer **130** based on one or more other criteria (e.g., upon completion of a flight by the aircraft **101**).

[0077] The EPGS **105** may be configured to identify fault events **121** according to severity level. For example, at **215**, the method **200** may include identifying, from among the fault events **121** found based on the search, fault events **121** of a target severity level. In an example, at **215**, the method **200** may include identifying fault events **121** of a lowest severity level.

[0078] At **220**, the method **200** may include counting the quantity of the fault events **121** of the target security level, for example, the lowest severity level. In an example, with reference to FIG. **1**, the EPGS **105** (e.g., using logic circuitry **115**-*b*) may identify that fault event **121**-*a*, fault event **121**-*b*, and fault event **121**-*d* (respectively included in flight leg data **140**-*a*, flight leg data **140**-*b*, and flight leg data **140**-*d*) are the same fault event **121** (same type of fault event **121**) (e.g., ‘Generator Low Oil Level’) and are all of the lowest severity level. In the example, the quantity of instances of the same fault event **121** (e.g., ‘Generator Low Oil Level’) is 3.

[0079] In some embodiments, the EPGS **105** (e.g., logic circuitry **115**-*b*) is configured to determine the quantity of instances of the same fault event **121** with respect to a target quantity of flights (e.g., the four most recent flights). For example, at **225**-*a*, the method **200** may include counting, with respect to flight leg data **140**-*a* through flight leg data **140**-*d* (e.g., four flights), a quantity of times that the same fault event **121** (e.g., ‘Generator Low Oil Level’) occurs. In the example, the EPGS **105** (e.g., using logic circuitry **115**-*b*) may identify that the same fault event **121** (e.g., ‘Generator Low Oil Level’) occurred three times over the past four flights corresponding to flight leg data **140**-*a* through flight leg data **140**-*d*.

[0080] In the example described herein, the flights include the four most recent flights. However, aspects of the present disclosure are not limited thereto, and the method **200** supports determining the quantity of instances of the same fault event **121** with respect to a target quantity of flights, for any time period. Non-limiting examples include the four most recent consecutive flights, every other flight (for a total of eight flights) during a target month, and the like.

[0081] Additionally, or alternatively, at **225**-*b*, the method **200** may include counting, with respect to the target time period associated with the search at **210**, a quantity of times that the same fault event **121** (e.g., ‘Generator Low Oil Level’) occurs. In some embodiments, the method **200** may include counting, with respect to any suitable time period (e.g., a time period different from the target time period associated with the search at **210**), a quantity of times that the same fault event **121** (e.g., ‘Generator Low Oil Level’) occurs.

[0082] At **235**, the method **200** may include promoting the severity of the target fault event **121** based on comparing (at **230**-*a* or at **230**-*b*) the quantity of the instances to a threshold quantity. In an example of promoting the severity of the target fault event **121**, the EPGS **105** is configured to generate an additional fault event **121** (e.g., a fault event **121**-*o*) associated with the same fault event **121** (e.g., fault event **121**-*a*, fault event **121**-*b*, and fault event **121**-*d*).

[0083] In the example, fault event **121**-*o* is of a higher severity level compared to the severity levels of fault event **121**-*a*, fault event **121**-*b*, and fault event **121**-*d*. In an example, the EPGS **105** (e.g., using logic circuitry **115**-*b*) is configured to assign the higher severity level to the fault event **121**-*o*.

[0084] In some embodiments, the aircraft system **102** may report the fault event **121**-*o* according to the higher severity level. For example, the aircraft system **102** may add the fault event **121**-*o* to the fault events **121**.

[0085] In some aspects, the aircraft system **102** (e.g., at EPGS **105** and/or at computing device **103**-*a*) may order the fault events **121** based on respective severity levels of the fault events **121** and display the fault events **121** based on the order. For example, each fault event **121** may be a database entry. The aircraft system **102** may report the fault event **121**-*o* as a new fault of the higher severity level, and the aircraft system **102** may add the fault event **121**-*o* to a data log including the fault events **121**, such that the fault event **121**-*o* is after fault event **121**-*n*.

[0086] In some embodiments, the aircraft system **102** is configured to generate respective database entries corresponding to recorded fault events **121** (e.g., fault event **121**-*a* through fault event **121**-*n*) and fault events **121** (e.g., fault event **121**-*o*) generated in accordance with the example techniques described herein. The aircraft system **102** is configured to store the database entries to database **104**.

[0087] In some embodiments, the aircraft system **102** is configured to generate timestamp data respectively corresponding to recorded fault events **121** (e.g., fault event **121**-*a* through fault event **121**-*n*) and fault events **121** (e.g., fault event **121**-*o*) generated in accordance with the example aspects of the present disclosure. In some aspects, the aircraft system **102** and/or the computing device **103**-*b* may display the fault events **121** based on respective timestamp data.

[0088] In some embodiments, the aircraft system **102** is configured to generate one or more corrective actions associated with a generated fault event **121** (e.g., fault event **121**-*o*). For example, in a case in which the fault event **121**-*o* is a ‘Persistent Generator Cooling System Oil Leak,’ the aircraft system **102** may generate corrective actions such as, for example, actions for troubleshooting and/or addressing the oil leak.

[0089] In some embodiments, the computing devices **103** (e.g., computing device **103**-*a*, computing device **103**-*b*) are configured to generate fault analysis data **145** associated with the fault event **121**-*o*. In some examples, the fault analysis data **145** may include corrective actions for addressing any of the fault events **121**. Aspects of the present disclosure support generating the fault analysis data **145** autonomously and/or semi-autonomously.

[0090] In some embodiments, the aircraft system **102** may transmit data or information described herein to the computing device **103**-*b*. In some other embodiments, the system **100** supports transferring (e.g., via wired or wireless data transmission) data or information described herein to the computing device **103**-*b* via electrical coupling to the NVM device **125** or electrical coupling to another suitable removable memory storage device on which data or information may be stored. For example, the system **100** supports data dumps of raw unfiltered data from the aircraft system **102** to the computing device **103**-*b*, based on which the computing device **103**-*b* may generate the fault analysis data **145**.

[0091] FIG. **3** illustrates an example flowchart of a method **300** that supports enhanced troubleshooting of aircraft EPGS fault events in accordance with one or more embodiments of the present disclosure. The method **300** may be implemented by aircraft system **102** and/or a computing device **103** described with reference to FIG. **1**.

[0092] At **305**, the method **300** includes continuously capturing, by one or more circuitry, status signals corresponding to a plurality of fault events, wherein the plurality of fault events are

associated with an electric power generating system of an aircraft.
[0093] At **310**, the method **300** includes counting, by the one or more circuitry, a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events.
[0094] At **315**, the method **300** includes generating, by the one or more circuitry, a second fault event associated with the fault event based on comparing the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level.
[0095] At **320**, the method **300** includes reporting, by the one or more circuitry, the second fault event according to the second severity level.
[0096] In some aspects, the method **300** may include continuously storing, by the one or more circuitry, the status signals to a circular buffer.
[0097] In some aspects, the method **300** may include classifying, by the one or more circuitry, the plurality of fault events according to respective severity levels of the plurality of fault events.
[0098] In some aspects, the method **300** may include adding, by the one or more circuitry, the second fault event to the plurality of fault events. In some aspects, the method **300** may include ordering, by the one or more circuitry, the plurality of fault events based on respective severity levels of the plurality of fault events. In some aspects, the method **300** may include displaying, by the one or more circuitry, the plurality of fault events based on the ordering.
[0099] In some aspects, the target severity level includes a lowest severity level. In some aspects, the second severity level is higher than the target severity level.
[0100] In the descriptions of the flowcharts herein, the operations may be performed in a different order than the order shown, or the operations may be performed in different orders or at different times. Certain operations may also be left out of the flowcharts, one or more operations may be repeated, or other operations may be added to the flowcharts.
[0101] Set forth are embodiments supported by the present disclosure.
[0102] Embodiment 1. A fault analysis system, comprising: one or more processors configured to: continuously capture status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; count a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; generate a second fault event associated with the fault event based on comparing the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and report the second fault event according to the second severity level.
[0103] Embodiment 2. The fault analysis system of Embodiment 1, further comprising: a memory device comprising a circular buffer, wherein the one or more processors are configured to continuously store the status signals to the circular buffer.
[0104] Embodiment 3. The fault analysis system of Embodiment 1 or Embodiment 2, wherein the one or more processors are configured to classify the plurality of fault events according to respective severity levels of the plurality of fault events.
[0105] Embodiment 4. The fault analysis system of any of Embodiment 1 through Embodiment 3, wherein the one or more processors are further configured to: add the second fault event to the plurality of fault events; order the plurality of fault events based on respective severity levels of the plurality of fault events; and display the plurality of fault events based on the order.
[0106] Embodiment 5. The fault analysis system of any of Embodiment 1 through Embodiment 4, wherein: the target severity level comprises a lowest severity level; and the second severity level is higher than the target severity level.
[0107] Embodiment 6. The fault analysis system of any of Embodiment 1 through Embodiment 5, further comprising: one or more rule based tables, wherein the one or more processors are configured to generate the second fault event based on the fault event and the one or more rule

based tables.

[0108] Embodiment 7. The fault analysis system of any of Embodiment 1 through Embodiment 6, wherein: the one or more processors are configured to assign the second severity level to the second fault event.

[0109] Embodiment 8. The fault analysis system of any of Embodiment 1 through Embodiment 7, wherein: the one or more processors are configured to generate a database entry corresponding to the fault event and store the database entry to a database; and the one or more processors are configured to generate a second database entry corresponding to the second fault event and store the second database entry to the database.

[0110] Embodiment 9. The fault analysis system of any of Embodiment 1 through Embodiment 8, wherein: the one or more processors are configured to generate first timestamp data associated with the fault event and second timestamp data associated with generation of the second fault event; and the one or more processors are configured to display the fault event and the second fault event based on the first timestamp data and the second timestamp data.

[0111] Embodiment 10. The fault analysis system of any of Embodiment 1 through Embodiment 9, wherein the one or more processors are further configured to:

[0112] generate one or more second corrective actions associated with the second fault event, wherein at least a portion of the one or more second corrective actions associated with the second fault event is different from one or more correction actions associated with the fault event.

[0113] Embodiment 11. The fault analysis system of any of Embodiment 1 through Embodiment 10, wherein the one or more processors are further configured to: determine a pattern associated with the instances of the fault event; and generate the second fault event based on comparing the pattern to a target pattern.

[0114] Embodiment 12. An apparatus, comprising: a controller including one or more processors and logic circuitry, wherein: processing circuitry of the one or more processors is configured to continuously capture status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; the logic circuitry is configured to count a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; the processing circuitry is configured to generate a second fault event associated with the fault event based on comparing, by the logic circuitry, the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and the processing circuitry is configured to report the second fault event according to the second severity level.

[0115] Embodiment 13. The apparatus of Embodiment 12, further comprising: a memory device comprising a circular buffer, wherein the processing circuitry is configured to continuously store the status signals to the circular buffer.

[0116] Embodiment 14. The apparatus of Embodiment 12 or Embodiment 13, wherein the logic circuitry is configured to classify the plurality of fault events according to respective severity levels of the plurality of fault events.

[0117] Embodiment 15. The apparatus of any of Embodiment 12 through Embodiment 14, wherein the processing circuitry is configured to: add the second fault event to the plurality of fault events; order the plurality of fault events based on respective severity levels of the plurality of fault events; and display the plurality of fault events based on the order.

[0118] Embodiment 16. A method comprising: continuously capturing, by one or more circuitry, status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; counting, by the one or more circuitry, a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; generating, by the one or more circuitry, a second fault event associated with the fault event based on comparing

the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and reporting, by the one or more circuitry, the second fault event according to the second severity level.
[0119] Embodiment 17. The method of Embodiment 16, further comprising: continuously storing, by the one or more circuitry, the status signals to a circular buffer.
[0120] Embodiment 18. The method of Embodiment 16 or 17, further comprising: classifying, by the one or more circuitry, the plurality of fault events according to respective severity levels of the plurality of fault events.
[0121] Embodiment 19. The method of any of Embodiment 16 through Embodiment 18, further comprising: adding, by the one or more circuitry, the second fault event to the plurality of fault events; ordering, by the one or more circuitry, the plurality of fault events based on respective severity levels of the plurality of fault events; and displaying, by the one or more circuitry, the plurality of fault events based on the ordering.
[0122] Embodiment 20. The method of any of Embodiment 16 through Embodiment 19, wherein: the target severity level comprises a lowest severity level; and the second severity level is higher than the target severity level.
[0123] The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.
[0124] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.
[0125] While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A fault analysis system, comprising: one or more processors configured to: continuously capture status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; count a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; generate a second fault event associated with the fault event based on comparing the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and report the second fault event according to the second severity level.
2. The fault analysis system of claim 1, further comprising: a memory device comprising a circular buffer, wherein the one or more processors are configured to continuously store the status signals to the circular buffer.
3. The fault analysis system of claim 1, wherein the one or more processors are configured to classify the plurality of fault events according to respective severity levels of the plurality of fault events.

2. The fault analysis system of claim 1, wherein the one or more processors are further configured to: add the second fault event to the plurality of fault events; order the plurality of fault events based on respective severity levels of the plurality of fault events; and display the plurality of fault events based on the order.
3. The fault analysis system of claim 1, wherein: the target severity level comprises a lowest severity level; and the second severity level is higher than the target severity level.
6. The fault analysis system of claim 1, further comprising: one or more rule based tables, wherein the one or more processors are configured to generate the second fault event based on the fault event and the one or more rule based tables.
4. The fault analysis system of claim 1, wherein: the one or more processors are configured to assign the second severity level to the second fault event.
5. The fault analysis system of claim 1, wherein: the one or more processors are configured to generate a database entry corresponding to the fault event and store the database entry to a database; and the one or more processors are configured to generate a second database entry corresponding to the second fault event and store the second database entry to the database.
6. The fault analysis system of claim 1, wherein: the one or more processors are configured to generate first timestamp data associated with the fault event and second timestamp data associated with generation of the second fault event; and the one or more processors are configured to display the fault event and the second fault event based on the first timestamp data and the second timestamp data.
10. The fault analysis system of claim 1, wherein the one or more processors are further configured to: generate one or more second corrective actions associated with the second fault event, wherein at least a portion of the one or more second corrective actions associated with the second fault event is different from one or more correction actions associated with the fault event.
7. The fault analysis system of claim 1, wherein the one or more processors are further configured to: determine a pattern associated with the instances of the fault event; and generate the second fault event based on comparing the pattern to a target pattern.
8. An apparatus, comprising: a controller comprising one or more processors and logic circuitry, wherein: processing circuitry of the one or more processors is configured to continuously capture status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; the logic circuitry is configured to count a quantity of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; the processing circuitry is configured to generate a second fault event associated with the fault event based on comparing, by the logic circuitry, the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and the processing circuitry is configured to report the second fault event according to the second severity level.
9. The apparatus of claim **12**, further comprising: a memory device comprising a circular buffer, wherein the processing circuitry is configured to continuously store the status signals to the circular buffer.
10. The apparatus of claim **12**, wherein the logic circuitry is configured to classify the plurality of fault events according to respective severity levels of the plurality of fault events.
11. The apparatus of claim **12**, wherein the processing circuitry is configured to: add the second fault event to the plurality of fault events; order the plurality of fault events based on respective severity levels of the plurality of fault events; and display the plurality of fault events based on the order.
12. A method comprising: continuously capturing, by one or more circuitry, status signals corresponding to a plurality of fault events, wherein the plurality of fault events are associated with an electric power generating system of an aircraft; counting, by the one or more circuitry, a quantity

of instances of a fault event of a target severity level with respect to a target quantity of flights, wherein the fault event is included in the plurality of fault events; generating, by the one or more circuitry, a second fault event associated with the fault event based on comparing the quantity of the instances to a threshold quantity, wherein the second fault event is of a second severity level different from the target severity level; and reporting, by the one or more circuitry, the second fault event according to the second severity level.
13. The method of claim **16**, further comprising: continuously storing, by the one or more circuitry, the status signals to a circular buffer.
18. The method of claim **17**, further comprising: classifying, by the one or more circuitry, the plurality of fault events according to respective severity levels of the plurality of fault events.
19. The method of claim **17**, further comprising: adding, by the one or more circuitry, the second fault event to the plurality of fault events; ordering, by the one or more circuitry, the plurality of fault events based on respective severity levels of the plurality of fault events; and displaying, by the one or more circuitry, the plurality of fault events based on the ordering.
20. The method of claim **17**, wherein: the target severity level comprises a lowest severity level; and the second severity level is higher than the target severity level.